

COMETS

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COMETS PRODUCE A STRIKING celestial spectacle when they enter the inner solar system. Their small nuclei become surrounded by a bright cloud, or coma, of dust and gas about 60,000 miles (100,000 km) across. Large comets that get close to the Sun also produce long, glowing tails that can extend many tens of millions of miles into space and are bright enough to be seen in Earth's sky.

ORBITS

Cometary orbits divide into two classes. Short-period comets orbit the Sun in the same direction as the planets. Most have orbital periods of about 7 years and get no farther from the Sun than Jupiter. Short-period comets were captured into the inner solar system by the gravitational influence of Jupiter. If they remain in these small orbits, they will decay quickly. Some, however, will be ejected by Jupiter into much larger orbits and then possibly recaptured. Intermediate- and long-period comets have orbital periods greater than 20 years (see pp.216–217). Their orbital planes are inclined at random to the plane of the solar system. Many of these comets travel huge distances into the interstellar regions. Most of the recorded comets get close to the Sun, where they develop comae and tails and can be easily discovered. There are vast numbers of comets on more distant orbits that are too faint to be found.

SPECTACULAR SIGHT

Most comets are only discovered when they are bright enough to glow in Earth's sky. Comet Hale-Bopp was discovered in late July 1997 and could be seen for several weeks afterward. It will return to Earth's sky again around 5400 CE.

